Chemiluminescence of Luminol

Purpose

To observe the light produced by the oxidation reaction of luminol and study and explain its chemical properties and reactions.

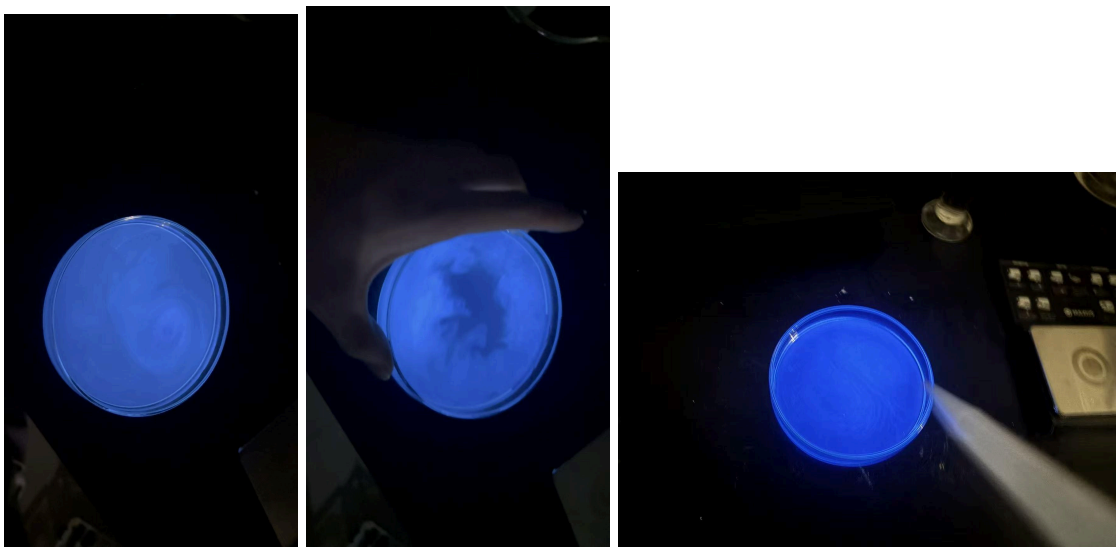
Materials

Chemical	Quantity
Luminol	0.010 g
Anhydrous sodium carbonate, Na_2CO_3	0.20 g
Sodium bicarbonate, $NaHCO_3$	1.20 g
Commercial ammonium carbonate / Baker's ammonia	0.025 g
Copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$	0.020 g
3% hydrogen peroxide, H_2O_2	2.5 mL
Distilled water	To prepare 50 mL of each solution

Procedure

- 30ml of distilled water was used to dissolve the sodium carbonate.
- Luminol added with sodium bicarbonate, ammonium carbonate and copper sulfate.
- 2.5 ml of 3% hydrogen peroxide was diluted with water to value of 50ml
- Mixing them together.

Observation

- A bright blue light was visible in the container
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Chemistry explanation

- The main reason for generating light during he reaction is that the excited product releases energy in the form of light once it returns to the ground state.
- The carbonate solution provides the alkaline condition in which luminol loses protons and forms a dianion. In the reaction, hydrogen peroxide acts as the oxidizing agent, and copper helps the reaction proceed faster.
- The oxidation involves peroxide intermediates that experience bond rearrangements, releasing nitrogen gas and producing 3- aminophthalate, and the energy comes from electronic excitation.
- This glow will weaken once the energy is all released; it is irreversible because the structure of the luminol has changed, with no reverse electron transfer.

Chemical Gardon

Purpose

To observe the growth of the chemical garden and explain chemical reactions such as osmosis and liquid flow.

Materials

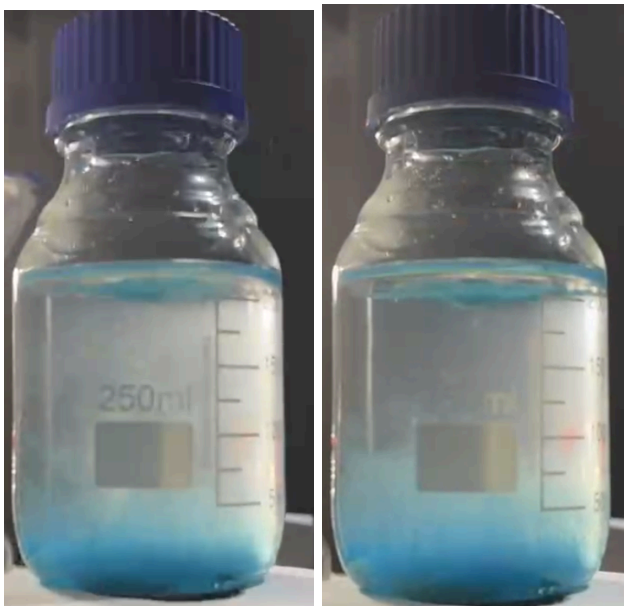
Chemicals or equipment	Quantity
Copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$	Small crystals
Sodium silicate solution	40% stock solution; dilution and volume
Distilled water	
Clear beaker or test tube	1
Laboratory spoon	1

Procedure

1. Add 150 mL of distilled water
2. Add 50 mL of 40% sodium silicate solution, ratio 1: 3.
3. Wait for 5 minutes until there are no bubbles.
4. Add 20 g of copper sulfate powder on the surface.
5. Waite for 30minnutes
6. The recorded video shows the slow growth

Observation

The blue “cloud” formed at the beginning, then dropped to the bottom. After 5 minutes, some tubes slowly grew from the crystal and bent with other branches at the middle parts,



Chemistry explanation

- Cooper sulfate dissolved and reacted with the water glass. Formed a thin membrane layer.
- Water enters the membrane by osmosis, which increases the pressure inside.
- This pressure breaks the membrane, which allows the copper liquid to flow out of the membrane.
- $Cu^{2+} + 2OH^{-} \rightarrow Cu(OH)_2$
- The liquid flows out of the membrane and reacts with the solution to build the hollow tubes
- It is vulnerable to mixing or shifting of the glass because the flow of liquid makes the chemical reaction that bends the tubes together.

Maillard Reaction in a Glucose- Glycine system

Purpose

To observe the color changes during the water-heating bath of glucose and glycine, to explain the chemical reactions different composition with control group.

Materials

Sample	Glucose	Glycine	0.1% (w/v) sodium carbonate solution	Final volume
A: Mixed reaction sample	1.60 g	0.64 g	2.00 mL	8.00 mL
B: Glucose control	1.60 g	None	2.00 mL	8.00 mL
C: Glycine control	None	0.64 g	2.00 mL	8.00 mL
D: Mixture without sodium carbonate	1.60 g	0.64 g	None	8.00 mL

Observation

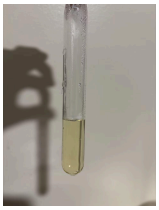
Sample A shows a deep brown color; it is the darkest solution among them. (figure1)

Sample B shows a light yellow color in the solution. Third darkest among the solutions. (figure2)

In sample C shows transparent and cololess mong the solutions.

In sample D, it is a dark brown color, after sample A.

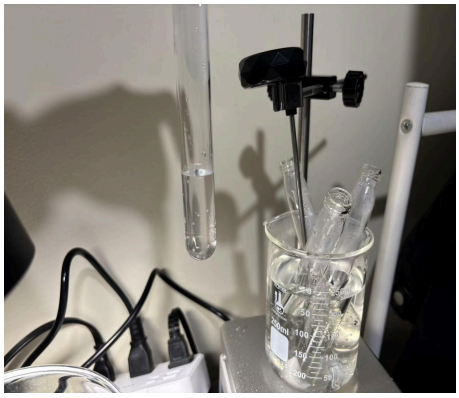

(figure1)


(figure2)


(figure3)


(figure4)





Chemical explanation

The Maillard reaction is a series of non-enzymatic reactions between glucose and the amino acid glycine. Heating facilitates the transformation of the intermediates into yellow and brown as the product of the reaction. Glucose is sometimes open-chain in water and has a CHO group at the end, which is reactive with the amino group of glycine and undergoes Amadori rearrangement; the reaction includes fragmentation and dehydration, causing the solution to absorb light and produce a dark color. The reason for the sample B is the color only result the degradation of sugar and sample C which lack of glucose cause the absence of the open chain tail CHO, where no reaction occurred. Sample D is that, even though there is no Sodium carbonate, the condition of the environment changed, causing less obvious color changes. Sodium carbonate makes the solution more alkaline: $CO_3^{2-} + H_2O \bullet HCO_3^- + OH^-$. This environment promotes sugar degradation nd easier for the reaction to occur.

Detection of peptide using the Biuret Test

Purpose

To observe the color of the reaction between tryptone, glycine, and water with alkaline copper. To understand how the chemical reaction of peptide structure affects the copper.

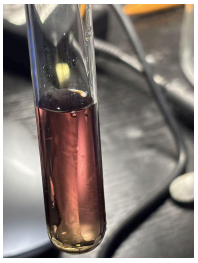
Material

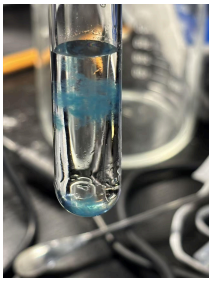
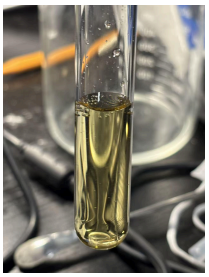
Tryptone, glycine, copper(II) sulfate solution, sodium hydroxide solution, distilled water, test tubes, a test-tube rack, pipettes

Procedure

- Add tryptone to A and D, Glycine to B, and distilled water to C
- Add sodium hydroxide solution to each tube.
- Add copper sulfate to A, B, and C. Add the same amount of water to D.

Observation

Tube	Composition	
A	Tryptone, NaOH, Copper Sulfate	

B	Glycine, NaOH, copper	
C	Water, NaOH, copper	
D	Tryptone, NaOH	

Chemical explanation

In an alkaline environment, the N and H of the peptide bond lose protons, causing the original nitrogen atom to pair with Cu2. Cu can form a coordinate bond with four nitrogen atoms, forming a chelate complex - there is a central ion that accepts the surrounding ligands' electrons to form a coordination bond (sharing two electrons). In this chelate complex, there will be two or more atoms and ligands connected. The deprotonated nitrogen atom becomes a strong ligand that affects the absorption of light. Tryptone itself has a yellow color. The color of A comes from the copper and peptide complex, B because it lacks a peptide bond, causes the combination of oxygen and copper forms copper glycinate, which is clear blue. C because there are no ligands, forms a precipitate. Therefore, the overall reaction comes from the coordination bonds between the surrounding ions and the central particle.

Formation of Turnbull’s blue

Purpose

To observe the reaction between iron sulfate and potassium ferricyanide and explain the chemical reaction, including the color and electron transfer.

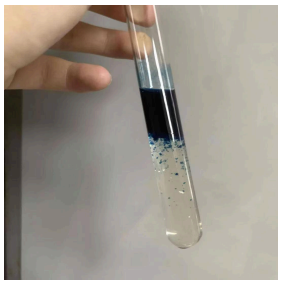
Materials

Iron sulfate solution, potassium ferricyanide solution

Procedure

Combine the iron sulfate and potassium ferricyanide solutions in a test tube.

Observation

Appears in different regions: dark blue at the top, small blue particles at the bottom.	

Chemical explanation

Potassium ferricyanide has a cyanide group that connects iron into a network; the electron transfer changes the iron oxidation states, producing Turnbull's blue, which contains both Fe^{2+} and Fe^{3+}; the dark blue region forms because of intervalence charge transfer. As the light moves electrons around the neighboring iron, which absorbs red light, it turns blue.

Magnesium with water using phenolphthalein

Purpose

To observe how magnesium reacts with water and use phenolphthalein to determine the change in alkalinity.

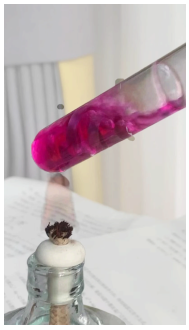
Materials

Magnesium, water, phenolphthalein solution. Alcohol lamp

Procedure

Magnesium is added to contact with water containing phenolphthalein, which appears in color and gas formation.

Observation

During heating, the solution appeared pink, magnesium ribbon still visible inside the tube.	

Chemical explanation

Magnesium reacts with water in an oxidation-reduction reaction. It loses electrons and becomes Mg2+, which water accepts the electrons. In the process, the transfer of electrons produces hydrogen gas and hydroxide ions that change the solution alkaline, this cause phenophthalein to lose protons changed the structure. The heating speed up the reaction.

Reduction of methylene blue by glucose

Purpose

To observe the repeated oxidation and reduction of methylene blue and explain the chemical reactions of glucose and oxygen.

Materials

Glucose, methylene blue solution, sodium hydroxide, distilled water.

Procedure

- Combine glucose, sodium hydroxide solution, and water
- Add Methylene blue solution
- Cap the bottle and wait for the liquid to return to a colorless state.
- Shake the bottle and observe the change in color.

Observations

Beginning	
Middle	

End It starts to fade out gradually (10 min)	

Chemical explanation

In an alkaline solution, glucose transfers electrons to methylene blue(C16H18N3S +H2O +2e – C16H19N3S+OH), reducing it to a colorless state. Shaking dissolves oxygen from the air, which reoxidizes it and restores the blue color (C6H12O6 +3OH– C6H11O7 +2H_2O + 2e).Methylene blue acts as the redox mediator, transferring electrons from glucose to oxygen.

Reduction of Copper II hydroxide by glucose

Prupose

To observe the oxidation-reduction reaction and color of the reducing ability of glucose in copper.

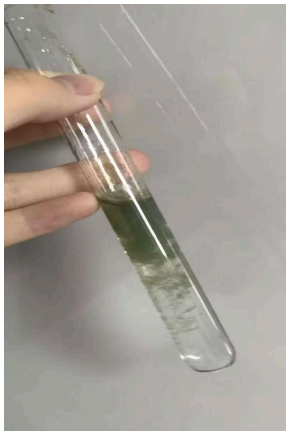
Materials

Glucose, copper II sulfate, sodium hydroxide, water

Procedure

- Combine copper II sulfate and sodium hydroxide.
- Add glucose

Observation

It appears darker green at the top of the test tube, with cloudy particles in the middle dividing the clear and dark regions.	

Chemical explanation

The glucose is commonly found in cyclic hemiacetal form in water, and only a small portion exists in an open-chain structure. In this rare case, the CHO is exposed. If it is consumed, the chemical bond equilibrium will shift and move along the open chain. In the Benedict reagent, Cu2 will be encapsulated by citric acid to form a soluble complex - (the charged center is a metal particle and the nearby molecules are arranged with

electrons) forming coordination bonds. In an alkaline environment, $CuOH_2$ will not form. When the aldehyde group is oxidized, it converts to a carboxylate, and the overall oxidation state of C1 rises. Cu^{2+} is reduced (KPU, Organic Chemistry). This insoluble precipitate will cause a color change as the sugar content increases. In this sample, Th below sugar content indicate the lower sugar had been put in.

Copper II and Ammonia Complex Formation

Purpose

To observe the reaction between ammonia and copper (II) nitrate and explain the chemical reaction between ammonia as a base that changes the color of copper precipitation.

Materials

Copper (II) nitrate, ammonia solution

Procedure

- Add 0.2 g copper (II) nitrate
- 10ml distilled water
- ~8 mL aqueous ammonia

Observation

A deep blue appears at the top region of the liquid, the pale blue remains at the bottom of the test tube, with solid material at the boundary.	

Chemical explanation

Ammonia is a weak base that produces hydroxide to form the pale blue color of copper (II) hydroxide. When there is enough ammonia, there will be more copper bound to ammonia, which causes the redissolving of the precipitate. The change in the coordination environment affects the light it absorbs; therefore, the solution turns blue at different levels. ($Cu2+ + 2OH- – Cu(OH)_2$) Because the density of the ammonia solution is relatively low, it will form a layer that floats on the upper layer, creating a situation where the concentrations alternate. In the middle layer, there is less ammonia and the weak base can only allow a small amount of ionization, allowing OH and Cu2 to combine. At the top layer, due to the excessive ammonia, NH3 can directly bind with Cu2 to form a dark blue phenomenon.

Crystallization of Sodium acetate - Hot ice

Purpose

To observe the reaction of supersaturated sodium acetate crystallization under the given conditions and understand the chemistry concept of triggering solidification of the liquid.

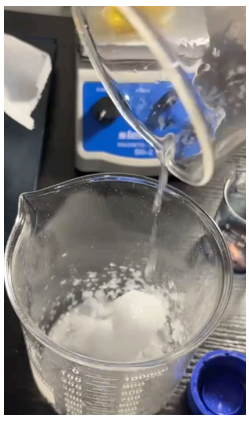
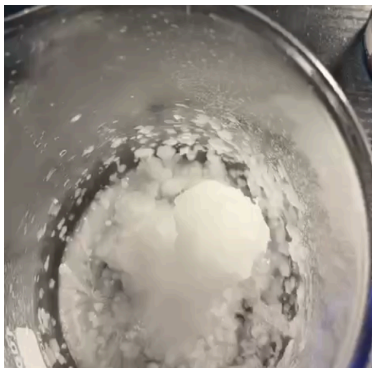
Materials

Sodium acetate trihydrate, distilled water, sodium acetate seed

Procedure

- 1g of sodium acetate trihydrate crystal reserved as seed.
- Add 125g sodium acetate trihydrate and 12.5mL of distilled water.
- It was heated to 80 degree for 15m and stirred until there is no more solid at the bottom.
- Place it on the table waite for cooling to room temperature ~ 22
- Place the seed at center of new container,
- The cooled solution is then pour on the crystal, observe the reaction.

Observation

White non-transparent crystal is formed as the liquid poured on the seed.	 

Chemical explanation

Once the sodium acetate trhydrate is heated, the solubility increases, in the experiment it completely dissolved at 80 degree form the solution, during the cooling it shouldn/t be any

disturbance because it may crystallize, due to it is supersaturated, it is likely crystallize or form solid particles, but in this case it is metastable which in a rest state.Nucleation reactions are necessary because we need a first seed particle cluster to ensure the crystallization of a liquid. When particles are added to the lattice, they release energy, causing each cluster to form a solid surface. During this process, a significant amount of energy is lost, and the liquid re-enters it. However, only clusters past a certain size can grow steadily and crystallize. The crystallization process and preparation require a clean container because the nucleation of sodium acetate requires a lot of energy. If there are residues in the container, it will disrupt this stability.The poured solution will blend with the existing crystals and form new crystals. During this process, it releases the latent heat energy from the supercooled liquid. The contact surfaces of the introduced liquids are different, and they grow randomly to form the final product.